Diversified applications of Computer Science and Engineering: Biology/bioinformatics, Artificial intelligence, Cryptography.

Recommended Texts:

1. A. Goel, *Computer Fundamentals*, 27th ed. Pearson, 2010
2. P. Norton, *Introduction To Computers*, 7th ed. McGraw-Hill, 2010
3. D. Reed, *A Balanced Introduction to Computer Science*, 3rd ed. Prentice Hall, 2010

CSE 4107 **Structured Programming I** **Credit: 3.00**

Introduction to Programming Concepts, Algorithm and Logic, Constants, Variables, Keywords and Data Types, Operators and expressions, Managing Input and Output Operations, Decision Making and Branching, Decision Making and Looping, Arrays, Multi-dimensional Arrays, Strings, User defined functions, Recursion, Structures and Unions, File Management in C, Pointers, Dynamic Memory Allocation and Linked List, The Preprocessor and some advanced topics, Advanced data types and operators.

Recommended Texts:

1. H. Schildt, *Teach Yourself C*, 3rd ed. McGraw-Hill, 1997
2. E. Balagurusamy, *Programming In Ansi C*, 8th ed. Noida, Uttar Pradesh, India: McGraw Hill Education, 2019

Hum 4142 **Arabic I** **Credit: 1.00**

Tajweed Rules of the Holy Quran; Letters and Pronunciation; Construction of words; Use of Numerical; Common Vocabularies; Name of Months, days and directions; Use of every day's conversation, dialogues and practice.

Recommended Texts:

1. M. Rashed, *Learn how to read Al-Qur'an*, `https://quranport.com/en/learn-how-to-read-quran/`, 2010

Hum 4144 **English I** **Credit: 1.00**

This course aims to give students of an international community accurate and meaningful communicating skills which will include expressions for personal identification (name, occupation, nationality etc.); body parts; time, day, week, months and years; daily program; education and future career; entertainment; travel; postal, telephonic